Project Title: Delhi weather AQI Dashboard and Analysis

Submitted in partial fulfillment of the requirements for the award of the degree of

Master of Computer Application (MCA)

By

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Enrollment No: O24MCA111082

Under the guidance of

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Certificate

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Declaration

I, **Shivam Singh**, hereby solemnly declare that the project report titled **"Delhi weather AQI Analysis and Visualization Dashboard"** submitted in partial fulfillment of the requirements for the award of the degree of **Master of Computer Application (MCA)** is my original work.

I further declare that:

- This project has been carried out by me during the academic year 2026 under the supervision of **Vandana Sivaraj**
- The work has not been submitted previously to any other university, institution, or examination body for the award of any degree, diploma, or certification.
- All sources of information used in this report have been duly acknowledged and referenced in accordance with academic ethics and plagiarism norms.
- The data presented in this report is authentic to the best of my knowledge and has not been fabricated or manipulated.
- I understand that if any part of this declaration is found to be false, my project may be rejected and disciplinary action may be taken as per institutional rules.

Place: CU university
Date: 15th may 2026

Student Signature: Shivam Singh

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Enrollment No: O24MCA111082

Acknowledgement

I would like to express my sincere gratitude to my project guide for their continuous support, valuable guidance, and encouragement throughout the development of this project. Their suggestions and feedback helped me improve my work at every stage.

I am also thankful to my department and faculty members for providing the necessary resources and knowledge required to complete this project successfully.

I would like to extend my thanks to my friends and classmates for their support and helpful discussions during the project work.

Finally, I am grateful to my family for their constant motivation and encouragement, which helped me stay focused and complete this project on time.

Project Overview:

This project focuses on analyzing and visualizing Delhi's weather conditions and Air Quality Index (AQI) data for the year 2025 using Python data analysis libraries. The goal is to understand weather trends, pollution patterns, and relationships between environmental factors through graphical visualizations and exploratory data analysis.

Objectives

- Analyze Delhi weather and AQI dataset
- Perform data cleaning and preprocessing
- Identify seasonal and monthly trends
- Visualize temperature, humidity, AQI, and pollution levels
- Understand correlations between weather conditions and air pollution

Dataset Features

The dataset includes:

- Date and time
- Temperature (°C)
- Humidity (%)
- Wind speed (kph)
- Pressure (mb)
- AQI Index
- PM2.5 and PM10 levels
- CO and NO2 pollutants
- Weather conditions
- Delhi locations

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook

    - Converted date columns into datetime format

2. Feature Engineering
    - Extracted Year, Month, and Day

- Created seasonal categories

3. Exploratory Data Analysis (EDA)
   Generated multiple graphs and charts to analyze:
   - Temperature distribution
   - AQI trends over months
   - Humidity vs Temperature
   - Wind speed distribution
   - Weather condition frequency
   - Correlation heatmaps
4. Visualization
   Used Matplotlib and Seaborn to create clear graphical insights.

Key Insights

- AQI levels are generally higher during winter months.
- PM2.5 and PM10 show strong correlation with AQI.
- Temperature and humidity have noticeable relationships.
- Seasonal weather changes significantly affect pollution levels.

Output

The project produces:

- Data visualization notebook
- Multiple graphs and charts

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Abstract

The “Delhi Weather & AQI Data Visualization” project aims to analyze and visualize weather conditions and air quality patterns in Delhi using a real-world dataset for the year 2025.

The project uses Python-based data analysis and visualization libraries such as Pandas, Matplotlib, and Seaborn to perform exploratory data analysis (EDA) on environmental data including temperature, humidity, wind speed, AQI index, PM2.5, PM10, CO, and NO2 levels.

The dataset is preprocessed by handling missing values, converting date formats, and creating additional features such as month, day, and seasonal categories. Various graphical visualizations such as histograms, line charts, scatter plots, bar charts, box plots, and heatmaps are generated to identify trends, correlations, and seasonal variations in weather and pollution levels.

The analysis helps in understanding how climatic conditions influence air quality in Delhi.

The project provides meaningful insights into pollution trends, temperature variations, and the relationship between atmospheric

conditions and AQI. This visualization-based approach makes environmental data easier to interpret and supports awareness and decision-making related to weather and air pollution monitoring.

Chapter 1 : Introduction

1.1 Background of Air Pollution

Air pollution has become one of the most serious environmental challenges faced by modern society. Rapid urbanization, industrial growth, increasing vehicular emissions, and expanding energy consumption have contributed significantly to the deterioration of air quality in many parts of the world. Over the past few decades, the concentration of harmful pollutants in the atmosphere has increased considerably, affecting both human health and the natural environment.

In developing countries, the problem of air pollution is particularly severe due to rapid population growth and unplanned urban expansion. Major metropolitan cities experience high levels of pollutants released from automobiles, industrial facilities, construction activities, and burning of fossil fuels. These pollutants accumulate in the atmosphere and lead to poor air quality conditions, especially in densely populated urban areas.

India is among the countries facing significant challenges related to air pollution. Several cities frequently record high pollution levels, which pose serious risks to public health. In recent years, environmental agencies and government bodies have taken initiatives to monitor and control air pollution by collecting and analyzing air quality data from different regions. However, the availability of large volumes of environmental data also creates the need for effective methods to analyze and interpret this information.

Air quality monitoring systems play an important role in understanding pollution patterns and identifying the sources of environmental degradation. These systems collect data related to various pollutants present in the atmosphere and provide valuable insights into the condition of the air in different geographical locations. The analysis of such data helps researchers, policymakers, and environmental organizations develop strategies for pollution control and environmental protection

1.2 Importance of Air Quality Monitoring

Air quality monitoring is an essential process for assessing the level of pollution present in the atmosphere. Monitoring systems measure the concentration of various pollutants such as particulate matter (PM2.5 and PM10), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), carbon monoxide (CO), and ozone (O_3). These pollutants originate from multiple sources including vehicle emissions, industrial processes, power generation plants, and natural events such as dust storms and forest fires.

The continuous monitoring of air quality allows environmental authorities to track pollution levels and detect any abnormal increases in pollutant concentrations. This information helps government agencies implement appropriate regulatory measures to reduce pollution and protect public health. Additionally, air quality monitoring provides essential data that can be used for research and environmental studies.

One of the key challenges in air quality monitoring is the effective interpretation of collected data. Large datasets generated from monitoring stations often require advanced analytical techniques to extract meaningful insights. Without proper analysis and visualization, raw environmental data may be difficult for decision-makers and the general public to understand.

Therefore, there is a growing need for systems that can analyze air quality data efficiently and present the results in a clear and interactive manner. Data analytics and visualization tools provide an effective solution for transforming complex datasets into meaningful information that can support environmental decision-making.

1.3 Air Quality Index (AQI)

The Air Quality Index (AQI) is a standardized indicator used to communicate the level of air pollution to the public. It provides a simplified representation of complex environmental data by converting pollutant concentrations into a single numerical value that indicates the overall air quality condition of a specific location.

The AQI scale typically ranges from values representing good air quality to levels that indicate hazardous pollution conditions. Different ranges of AQI values correspond to different health categories, helping individuals understand how air pollution may affect their health. For example, lower AQI values generally indicate clean air with minimal health risks, while higher values suggest unhealthy air conditions that may cause respiratory problems and other health issues.

Environmental protection agencies around the world use AQI as a standard method to inform the public about daily air quality conditions. Governments often publish AQI reports to help people take necessary precautions during periods of high pollution. For instance, individuals with respiratory diseases may avoid outdoor activities when the AQI level indicates unhealthy conditions.

Because AQI combines multiple pollutant measurements into a single index, it provides a convenient way to compare air quality levels across different cities and time periods. This makes AQI an important tool for environmental analysis and public awareness.

1.4 Need for Data Analysis and Visualization

With the advancement of technology, large volumes of environmental data are being generated from monitoring stations, satellites, and research institutions. Analysing this data manually is both time-consuming and inefficient. As a result, data analytics techniques have become increasingly important in environmental studies.

Data analysis helps researchers identify patterns, trends, and relationships within air quality datasets. By analyzing historical data, it becomes possible to understand how pollution levels change over time and how different factors influence air quality conditions.

However, raw data tables alone are often difficult to interpret. Visualization techniques such as charts, graphs, and dashboards allow complex information to be presented in a more understandable form. Interactive dashboards enable users to explore datasets dynamically and gain insights into environmental trends without requiring advanced technical knowledge.

Visualization tools also play a significant role in communicating environmental information to policymakers and the general public. Clear visual representations of air quality trends can support informed decision-making and promote awareness about pollution issues.

1.5 Role of Data Analytics in Environmental Monitoring

Advancements in technology have led to the generation of large volumes of environmental data from monitoring stations, satellites, and research institutions. Analyzing this data manually is both time-consuming and inefficient. As a result, data analytics has become an essential component of modern environmental research.

Data analytics techniques allow researchers to process large datasets and extract meaningful insights. By analyzing historical air quality data, it becomes possible to identify patterns, detect anomalies, and understand the factors that influence pollution levels. These insights help researchers develop models that can predict future pollution trends and evaluate the effectiveness of environmental policies.

In addition to traditional statistical methods, modern data analytics tools enable interactive exploration of datasets. Visualization techniques such as charts, graphs, and dashboards make it easier to interpret complex information and communicate findings to a wider audience. These tools transform raw data into visual representations that highlight key trends and relationships within the data.

1.6 Need for Data Visualization and Dashboards

Data visualization plays a critical role in transforming complex datasets into understandable information. While raw data tables may contain valuable information, they are often difficult to interpret without visual representation. Charts, graphs, and dashboards provide a more intuitive way of analyzing data and identifying patterns.

Interactive dashboards have become widely used in various fields such as business analytics, healthcare, finance, and environmental monitoring. These dashboards allow users to explore datasets dynamically, apply filters, and view data from different perspectives. By presenting information in a visually appealing format, dashboards improve the accessibility and usability of data.

In the context of environmental monitoring, visualization dashboards can help researchers and policymakers quickly identify pollution trends, compare air quality across different cities, and evaluate the effectiveness of pollution control measures. Such systems also improve public awareness by presenting environmental information in a clear and understandable way.

1.7 Motivation for the Project

The increasing concern about air pollution and its impact on public health has motivated the development of tools that can effectively analyse and visualize environmental data. While large datasets related to air quality are available, there is often a lack of user-friendly systems that allow individuals to explore this data interactively.

The motivation behind this project is to develop a system that integrates data analysis, database management, and visualization techniques to provide meaningful insights into air quality trends. By creating an interactive dashboard, users can easily analyse pollution data, compare air quality levels across cities, and understand how pollution patterns change over time.

Such a system can support environmental awareness and help individuals gain a better understanding of air quality conditions in different regions.

1.8 Problem Statement

Despite the availability of air quality datasets, analyzing and interpreting this data can be challenging without appropriate tools. Large datasets stored in databases often require technical expertise to query and analyse, making it difficult for non-technical users to extract meaningful insights.

Furthermore, many traditional reporting systems rely on static data presentations that do not allow users to interact with the data. This limits the ability to explore different aspects of air quality trends and compare pollution levels across locations.

Therefore, there is a need for an integrated system that can process air quality data efficiently and present the results through interactive visualizations and dashboards.

1.9 Objectives of the Project

The main objectives of this project are:

- To analyze air quality data collected from different cities.
- To store and manage environmental data using a relational database system.
- To develop data processing and analysis modules using Python.
- To create interactive visualizations that display AQI trends and pollutant levels.
- To design a user-friendly dashboard interface that allows users to explore air quality data easily.

1.10 Scope of the Project

- The scope of this project focuses on the analysis and visualization of air quality data using modern data analytics tools. The system integrates multiple technologies, including Python for data processing, MySQL for database management, and Power BI for advanced data visualization.
- In addition, a web-based interface is developed to allow users to interact with the dataset and view graphical representations of pollution trends. The system enables users to select different cities and analyse AQI trends through dynamic charts and tables.
- Although the project focuses on selected cities and datasets, the system design can be extended to include additional data sources and more advanced analytical features in the future.

1.11 Organization of the Report

This project report is organized into several chapters to provide a clear understanding of the work carried out in this study.

Chapter 1 introduces the background of air pollution, the importance of air quality monitoring, and the motivation behind the project.

Chapter 2 presents the literature review, which discusses previous research and studies related to air quality analysis and visualization.

Chapter 3 describes the system design and architecture of the proposed solution.

Chapter 4 explains the implementation process, including the technologies and tools used for developing the system.

Chapter 5 presents the results and analysis obtained from the developed dashboard.

Finally, Chapter 6 summarizes the conclusions of the project and suggests possible future improvements.

Chapter 2: System Study

2.1 Existing System

In the present time, air quality monitoring is mainly done through government websites and a few online platforms. These systems provide Air Quality Index (AQI) data for different cities. However, most of these platforms are not very user-friendly and are often difficult to understand for general users.

Many existing systems display data in a static format, such as tables or simple charts, without much interaction. Users cannot easily filter data based on their needs, such as selecting a specific city or analyzing trends over time. Also, some websites are overloaded with information, making it confusing for beginners.

Another issue is that many systems do not provide clear visualization of data. While the data is available, it is not presented in a way that helps users quickly understand air quality conditions.

2.2 Problems in Existing System

There are several limitations in the existing systems:

- Most platforms lack interactive features for users
- Data is often displayed in a complex format
- Difficult to compare AQI values between cities
- Lack of proper graphical representation
- Not suitable for beginners or non-technical users
- Limited customization options
- Slow or cluttered user interfaces

Because of these issues, users may find it difficult to understand and analyze air quality data effectively.

2.3 Proposed System

To overcome the limitations of the existing system, this project proposes an **Air Quality Analysis and Visualization Dashboard**.

This system is a web-based application where users can select a city and view its AQI data in both table and graphical formats. In addition to single city analysis, the system also provides a comparison feature that allows users to select two cities and analyze their AQI data together. This helps users to better understand differences in pollution levels. The main aim of this system is to make air quality data easy to understand and accessible to everyone.

The system allows users to:

- Select a city from a dropdown menu
- View AQI data in a structured table
- Visualize AQI trends using graphs
- Compare AQI levels between two cities
- View average AQI values for selected cities
- Analyze pollution differences using graphical comparison

This makes the analysis simple and more interactive compared to existing systems.

2.4 Advantages of Proposed System

The proposed system has several advantages:

- Simple and user-friendly interface
- Interactive data visualization using graphs
- Easy selection of cities
- Faster data access and display
- Helps users understand AQI trends clearly
- Lightweight and efficient system

This system improves the overall user experience and makes data analysis easier.

2.5 Feasibility Study

a) Technical Feasibility

The project is technically feasible as it uses widely available technologies such as Python, Flask, MySQL, Pandas, and Plotly. These technologies are easy to use, well-supported, and suitable for developing data-driven web applications. The system also supports additional functionalities such as comparison between two cities and calculation of average AQI, which can be efficiently handled using these tools..

b) Economic Feasibility

This project does not require any expensive tools or software. All technologies used are free or open-source, making it cost-effective. The system can be developed and deployed with minimal resources.

c) Operational Feasibility

The system is easy to operate and does not require any special training. Users can select a single city or compare two cities to view AQI data, graphs, and average pollution levels. The simple interface ensures smooth user interaction and ease of use.

2.6 Technologies Used

Python

Python is used as the main programming language for the project. It helps in data processing and backend development.

MySQL

MySQL is used as the database to store AQI data for different cities.

Pandas

Pandas is used for handling and processing data efficiently.

Plotly

Plotly is used to create interactive graphs for data visualization.

Power BI

Power BI is used to design interactive dashboard of AQI Analysis.

2.7 System Requirements

Hardware Requirements

- Computer or laptop
- Minimum 4GB RAM

Software Requirements

- Python
- MySQL Database
- Code Editor (VS Code / PyCharm)
- Web Browser (Chrome, Edge, etc.)

2.8 System Architecture and Working

The system works in a simple and structured way:

1. The user opens the web application
2. The user logs in using valid credentials
3. The user selects either a single city or two cities for comparison from the dropdown menu
4. The request is sent to the Flask server
5. The server fetches AQI data from the MySQL database based on the selected input
6. Data is processed using Pandas (filtering, sorting, and average calculation)
7. A graph is generated using Plotly (single-city trend or comparison graph)
8. The system displays:

• AQI data in table format (single or side-by-side tables)

• Graphical visualization

• Average AQI values (in case of comparison)

9. The results are displayed on the dashboard

This workflow ensures smooth functioning, efficient data processing, and an interactive user experience.

Chapter 3: System Analysis

1. Introduction

System analysis is the process of studying, understanding, and evaluating the requirements, functionality, and workflow of a system before implementation. In the Delhi Weather & AQI Analysis System, system analysis was performed to understand how environmental data can be collected, processed, analyzed, and visualized efficiently.

The analysis focused on:

- Data flow
- Functional requirements
- User requirements
- System modules
- Database operations
- Dashboard functionality

The primary objective of the system is to monitor and analyze Delhi's weather conditions and air quality using data analytics and visualization techniques.

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2. Existing System Analysis

Existing System

Traditional environmental monitoring systems mainly rely on:

- Manual pollution reports
- Static government records
- Non-interactive monitoring systems

Problems in Existing System

- Lack of real-time visualization
- Limited analytical capabilities
- Difficult interpretation of large datasets
- No centralized dashboard
- Slow environmental analysis process

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3. Proposed System Analysis

Proposed System

The proposed system introduces an automated environmental analysis platform using:

- Python
- SQL
- Power BI

- Data visualization techniques

Features of Proposed System

- Automated data processing
- AQI trend analysis
- Interactive dashboard visualization
- Pollution hotspot identification
- Weather and pollution correlation analysis

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4. Objectives of the System

The major objectives are:

- Analyze weather and AQI data
- Identify pollution trends
- Generate graphical visualizations
- Monitor pollutant levels
- Support environmental awareness
- Provide analytical insights using dashboards

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5. Functional Requirements

A. Data Collection Module

The system should:

- Import CSV datasets
- Read weather and AQI records
- Store environmental data

B. Data Preprocessing Module

The system should:

- Handle missing values
- Remove duplicate records
- Convert date formats
- Create additional analytical features

C. Database Management Module

The system should:

- Store structured AQI data
- Execute SQL queries
- Retrieve analytical outputs

D. Analysis Module

The system should:

- Calculate average AQI
- Detect pollution hotspots
- Analyze pollutant contributions
- Perform correlation analysis

E. Visualization Module

The system should:

- Generate graphs and charts
- Display dashboard KPIs
- Provide interactive reports

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6. Non-Functional Requirements

Performance

- Fast query execution
- Efficient data processing

Reliability

- Accurate analytical outputs
- Stable dashboard operation

Scalability

- Support future dataset expansion
- Handle additional monitoring stations

Usability

- Easy-to-use dashboard interface
- Simple graphical interpretation

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7. Feasibility Analysis

A. Technical Feasibility

The project is technically feasible because:

- Python libraries are freely available
- SQL databases support analytical operations
- Power BI provides effective dashboard tools

B. Economic Feasibility

The system is cost-effective because:

- Open-source technologies are used
- Minimal hardware requirements
- No expensive infrastructure required

C. Operational Feasibility

The system is operationally feasible because:

- Users can easily understand dashboards
- Visualization simplifies environmental analysis
- Data processing is automated

---

8. System Workflow Analysis

Workflow

```
CSV Dataset
↓
Data Cleaning & Preprocessing
↓
Feature Engineering
↓
SQL Database Storage
↓
SQL Query Analysis
↓
Python Visualization
↓
Power BI Dashboard
↓
Result Interpretation
```

---

9. Input Analysis

Input Data

The system accepts:

- Weather records
- AQI values
- Pollutant concentrations
- Wind speed data
- Temperature and humidity values

Input Format

- CSV file format

---

10. Process Analysis

The processing phase includes:

- Data cleaning
- Statistical analysis
- SQL computations
- Correlation analysis
- Trend generation

Processing Tools

- Pandas
- NumPy
- SQL Queries
- Matplotlib
- Seaborn

---

11. Output Analysis

The system generates:

- AQI dashboards
- Pollution trend graphs
- KPI summaries
- Correlation heatmaps
- Pollution hotspot reports

Output Benefits

- Easy interpretation of pollution data
- Better environmental understanding
- Improved analytical reporting

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12. System Constraints

The current system has some limitations:

- Uses static historical dataset
- No live AQI monitoring
- Limited predictive analysis
- Requires manual CSV updates

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13. Advantages of the System

- Automated environmental analysis
- Interactive dashboard visualization
- Efficient SQL-based reporting
- Improved pollution monitoring
- Easy graphical representation
- Supports environmental research

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14. Future Enhancement Possibilities

The system can be extended using:

- Real-time APIs
- Machine learning models
- IoT sensor integration
- Geographic heatmaps
- Mobile applications
- AI-based AQI prediction systems

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15. Conclusion

The system analysis of the Delhi Weather & AQI Analysis System demonstrates that the project provides an effective framework for environmental monitoring and data visualization. The system efficiently processes AQI and weather data, performs analytical operations, and generates meaningful graphical insights using Python, SQL, and Power BI technologies.

The analysis confirms that the proposed system is technically feasible, cost-effective, user-friendly, and suitable for environmental data analysis and pollution monitoring applications.

Chapter 4: System Design

1. Introduction

The Delhi Weather & AQI Analysis System is designed to collect, store, process, analyze, and visualize environmental data related to weather conditions and air pollution in Delhi. The system integrates CSV datasets, SQL database management, Python-based data analysis, and dashboard visualization tools to generate meaningful insights about AQI trends and weather patterns.

The system helps users:

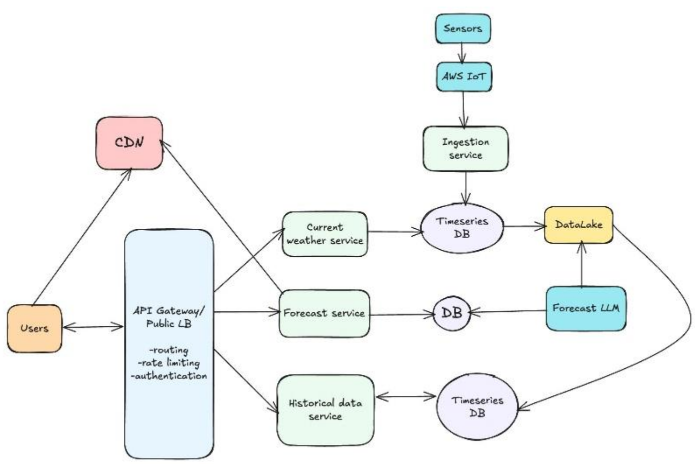
- Monitor pollution levels
- Analyze weather conditions
- Identify pollution hotspots
- Visualize environmental trends
- Support environmental decision-making

---

2. System Architecture

Architecture Flow

```
+-------------------+
|  Data Source      |
| (CSV Dataset)     |
+-------------------+
|
v
+-------------------+
| Data Preprocessing|
| Python + Pandas   |
+-------------------+
|
v
+-------------------+
| SQL Database      |
| PostgreSQL/MySQL  |
+-------------------+
|
v
+-------------------+
| Data Analysis     |
| SQL Queries       |
+-------------------+
|
v
+-------------------+
| Visualization     |
| Power BI / Python|
+-------------------+
|
v
+-------------------+
```

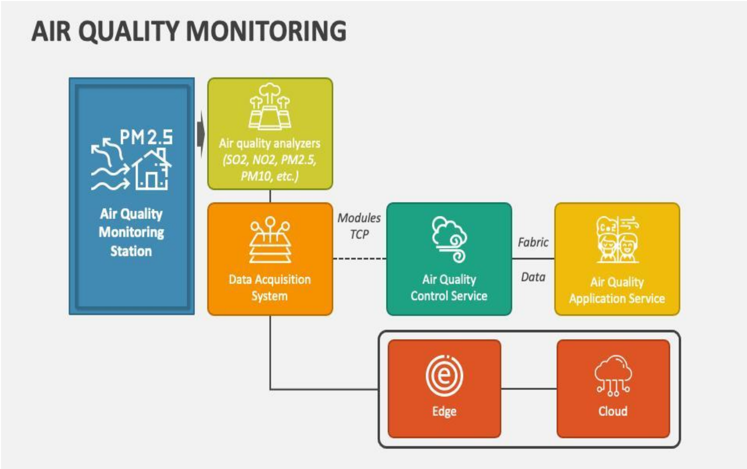


```
| Dashboard Output |
+------------------+
```

3. System Components

A. Data Source Layer

The system uses a CSV dataset containing:



- AQI values
- PM2.5 and PM10 data
- Temperature
- Humidity
- Wind speed
- CO and NO2 levels
- Time and location information

Input File

- `delhi-weather-aqi-2025.csv`

B. Data Preprocessing Layer

Technologies Used

- Python
- Pandas
- NumPy

Functions

- Handle missing values
- Remove duplicate records
- Convert date formats
- Feature engineering
- Categorize seasons and wind levels

Output

Cleaned and structured environmental dataset.

C. Database Layer

Database Used

- PostgreSQL / MySQL

Main Table

```
CREATE TABLE aqi (
Date_IST DATE,
Location_Delhi VARCHAR(100),
AQI_index FLOAT,
PM2_5 FLOAT,
PM10 FLOAT,
CO FLOAT,
NO2 FLOAT,
Temp_C FLOAT,
Humidity FLOAT,
Windspeed_kph FLOAT
);
```

Responsibilities

- Store environmental data
- Perform analytical queries
- Manage structured records
- Support dashboard integration

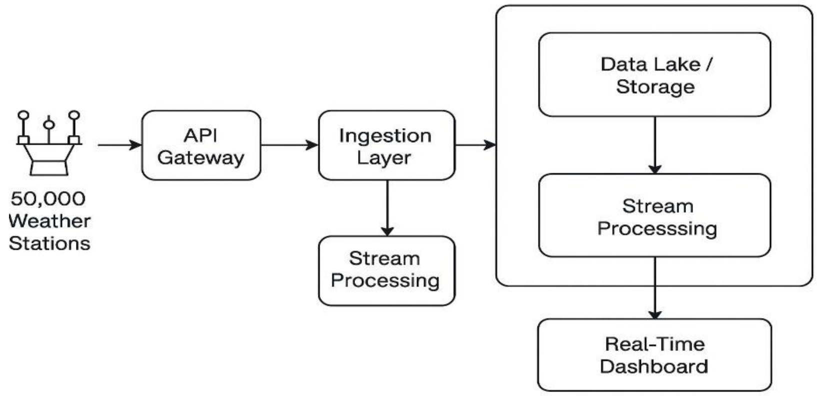
---

4. Analysis Layer

SQL-Based Analytics

The analysis module executes queries to:

- Calculate average AQI
- Detect pollution hotspots
- Analyze pollutant contributions
- Study wind speed impact
- Generate trend analysis



Example Query

```
SELECT Location_Delhi,
AVG(AQI_index) AS Avg_AQI
FROM aqi
GROUP BY Location_Delhi
ORDER BY Avg_AQI DESC;
```

---

5. Visualization Layer

Tools Used

- Power BI
- Matplotlib
- Seaborn

Dashboard Features

- KPI Cards
- Bar Charts

- Line Graphs
- Donut Charts
- Scatter Plots
- Heatmaps

Key Visual Insights

- AQI trends by location
- Pollution by wind category
- Pollutant contribution analysis
- Hourly AQI monitoring

---

6. Functional Modules

Module 1 – Data Collection

Collects weather and AQI data from CSV files or APIs.

Module 2 – Data Cleaning

Processes raw data into usable format.

Module 3 – Database Management

Stores and retrieves environmental records.

Module 4 – Data Analysis

Performs SQL analytics and statistical operations.

Module 5 – Dashboard Visualization

Displays interactive reports and graphs.

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7. Advantages of the System

- Real-time environmental monitoring capability
- Easy visualization of pollution trends
- SQL-based analytical reporting
- Supports decision-making for pollution control
- Scalable for future machine learning integration

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8. Limitations

- Uses static CSV dataset
- No live API integration
- Limited predictive analytics
- Dashboard depends on available historical data

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9. Future Enhancements

The system can be improved by adding:

- Real-time AQI APIs
- Machine learning prediction models
- IoT sensor integration
- Mobile application support
- GIS-based pollution heatmaps
- Automated alert systems

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10. Conclusion

The Delhi Weather & AQI Analysis System provides a complete framework for environmental data monitoring and visualization. By integrating Python, SQL, and dashboard technologies, the system efficiently analyzes weather and pollution data to identify trends, pollution hotspots, and environmental risks. The project demonstrates practical applications of data analytics, visualization, and business intelligence in environmental monitoring systems.

chapter 5: System Implémentation

1. Introduction

The implementation of the Delhi Weather & AQI Analysis System was carried out using Python, SQL, and Power BI technologies. The system was developed to process environmental data, perform pollution analysis, and visualize AQI trends through an interactive dashboard.

The implementation process includes:

- Data collection
- Data preprocessing
- Database creation
- SQL query execution
- Data visualization
- Dashboard development

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2. Development Environment

Software Tools Used

Tool	Purpose
Python	Data preprocessing and analysis
Pandas	Data manipulation
NumPy	Numerical computations
PostgreSQL/MySQL	Database management
SQL	Data querying
Jupyter Notebook	Analysis environment
Power BI	Dashboard visualization
Matplotlib & Seaborn	Graph plotting

---

3. Dataset Implementation

Dataset Used

The system uses the CSV dataset:

```
delhi-weather-aqi-2025.csv
```

Dataset Attributes

- Date and Time
- AQI Index
- PM2.5
- PM10

- CO
- NO2
- Temperature
- Humidity
- Wind Speed
- Location Details

---

4. Data Preprocessing Implementation

Step 1 – Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Purpose

Required libraries are imported for data handling and visualization.

Step 2 – Load Dataset

```
df = pd.read_csv("delhi-weather-aqi-2025.csv")
```

Purpose

Loads the environmental dataset into a Pandas DataFrame.

Step 3 – Handle Missing Values

```
df.isnull().sum()
df.dropna(inplace=True)
```

Purpose

Removes incomplete records from the dataset.

Step 4 – Convert Date Format

```
df['date_ist'] = pd.to_datetime(df['date_ist'])
```

Purpose

Converts the date column into datetime format for time-based analysis.

Step 5 – Feature Engineering

```
df['Month'] = df['date_ist'].dt.month
df['Year'] = df['date_ist'].dt.year
```

Purpose

Creates additional features for monthly and yearly analysis.

5. Database Implementation

Step 1 – Create Database

```
CREATE DATABASE weather_aqi;
```

Step 2 – Create AQI Table

```
CREATE TABLE aqi (
date_ist DATE,
location_delhi VARCHAR(100),
aqi_index FLOAT,
pm2_5 FLOAT,
pm10 FLOAT,
co FLOAT,
no2 FLOAT,
temp_c FLOAT,
humidity FLOAT,
windspeed_kph FLOAT
);
```

Purpose

Stores structured environmental and pollution data.

Step 3 – Import CSV Data

```
COPY aqi
FROM 'delhi-weather-aqi-2025.csv'
DELIMITER ','
CSV HEADER;
```

Purpose

Imports dataset records into the SQL database.

6. SQL Query Implementation

Average AQI Calculation

```
SELECT AVG(aqi_index) AS average_aqi
FROM aqi;
```

Output

Calculates overall air quality level in Delhi.

Pollution Hotspot Detection

```
SELECT location_delhi,
AVG(aqi_index) AS avg_aqi
FROM aqi
GROUP BY location_delhi
ORDER BY avg_aqi DESC;
```

Output

Identifies highly polluted areas.

Wind Category Analysis

```
SELECT
CASE
WHEN windspeed_kph < 5 THEN 'Low Wind'
WHEN windspeed_kph BETWEEN 5 AND 10 THEN 'Medium Wind'
ELSE 'High Wind'
END AS wind_category,
AVG(aqi_index) AS avg_aqi
FROM aqi
GROUP BY wind_category;
```

Output

Analyzes AQI variation based on wind conditions.

---

7. Visualization Implementation

Temperature Distribution Graph

```
sns.histplot(df['temp_c'], bins=30, kde=True)
plt.title("Temperature Distribution")
plt.show()
```

AQI Trend Analysis

```
monthly_aqi = df.groupby('Month')['aqi_index'].mean()

plt.plot(monthly_aqi.index, monthly_aqi.values)
plt.title("Monthly AQI Trend")
plt.show()
```

Correlation Heatmap

```
sns.heatmap(df.corr(), annot=True)
plt.show()
```

Purpose

Displays relationships between pollutants and weather parameters.

---

8. Dashboard Implementation

Dashboard Tool

- Power BI Desktop

Dashboard Components

- KPI Cards
- Bar Charts

- Donut Charts
- Line Charts
- Scatter Plots

Dashboard Workflow

```
CSV Dataset
↓
Python Cleaning
↓
SQL Database
↓
SQL Queries
↓
Power BI Dashboard
```

---

9. Testing and Validation

Validation Performed

- Checked missing values
- Verified SQL query outputs
- Compared dashboard values with dataset
- Tested graph generation

Result

The system successfully generated:

- AQI statistics
- Pollution trend analysis
- Interactive dashboard visualizations

---

10. Performance Evaluation

The system efficiently handled:

- Large environmental datasets
- SQL analytical operations
- Dashboard rendering
- Visualization generation

The implementation provides fast analysis and meaningful graphical insights.

---

11. Conclusion

The implementation of the Delhi Weather & AQI Analysis System successfully integrates Python, SQL, and Power BI technologies to analyze environmental pollution data. The system processes raw weather and AQI data, performs analytical computations, and generates visual dashboards for easy interpretation of pollution trends and environmental conditions.

The project demonstrates practical implementation of:

- Data preprocessing
- Database management
- SQL analytics
- Dashboard visualization
- Environmental data analysis

The developed system can be further extended for real-time monitoring and predictive analytics applications.

Chapter 6: Testing

1. Introduction

Testing is an important phase in the development of the Delhi Weather & AQI Analysis System. It ensures that the system functions correctly, produces accurate analytical results, and generates reliable visualizations from the environmental dataset.

The testing process was conducted on:

- Data preprocessing modules
- SQL database operations
- Analytical queries
- Visualization outputs
- Dashboard performance

The objective of testing was to verify system accuracy, consistency, reliability, and performance.

---

2. Testing Objectives

The main objectives of testing are:

- Verify correct dataset loading
- Ensure accurate data preprocessing
- Validate SQL query results
- Check dashboard visualizations
- Detect errors and inconsistencies
- Improve system reliability

---

3. Types of Testing Performed

A. Unit Testing

Unit testing was performed on individual modules such as:

- CSV loading
- Missing value handling
- Date conversion
- SQL queries

- Graph generation

Example

```
df.isnull().sum()
```

Result

Successfully identified missing values in the dataset.

B. Integration Testing

Integration testing verified communication between:

- Python preprocessing
- SQL database
- Power BI dashboard

Result

All modules exchanged data successfully without errors.

C. Database Testing

Database testing ensured:

- Table creation accuracy
- Correct CSV import
- Proper query execution
- Data consistency

Example Query

```
SELECT COUNT(*) FROM aqi;
```

Result

Confirmed successful insertion of all dataset records.

D. Visualization Testing

Graphs and dashboard components were tested to ensure:

- Proper chart rendering
- Correct labels and titles
- Accurate KPI calculations
- Consistent data representation

Tested Charts

- AQI trend graph
- Temperature distribution
- Wind speed analysis
- Correlation heatmap
- Donut chart visualization

Result

All visualizations displayed correct analytical insights.

E. Performance Testing

Performance testing measured:

- Query execution speed
- Dashboard loading performance
- Dataset processing efficiency

Result

The system handled large datasets efficiently with smooth dashboard performance.

---

4. Test Cases

Test Case ID	Test Description	Expected Result	Status
TC01	Load CSV dataset	Dataset loads successfully	Passed
TC02	Check missing values	Missing values identified	Passed
TC03	Convert date column	Correct datetime conversion	Passed
TC04	Create SQL table	Table created successfully	Passed
TC05	Import dataset into database	Records inserted correctly	Passed
TC06	Execute AQI queries	Accurate results generated	Passed
TC07	Generate graphs	Charts displayed properly	Passed
TC08	Dashboard KPI validation	KPI values matched dataset	Passed
TC09	Correlation heatmap generation	Heatmap generated successfully	Passed
TC10	Dashboard loading	Dashboard loaded without errors	Passed

---

5. Error Handling

The system was tested for possible errors such as:

- Missing CSV values
- Incorrect date formats
- SQL syntax errors

- Visualization rendering issues

Handling Methods

- Null value checking
- Data type conversion
- Query validation
- Exception handling in Python

---

6. Validation of Dashboard Results

The Power BI dashboard values were compared with:

- SQL query outputs
- Python analytical results
- Dataset statistics

Validation Result

Dashboard KPIs and charts accurately represented the environmental data.

---

7. Security and Reliability Testing

The system was tested for:

- Data consistency
- Query reliability
- Stable dashboard performance

Result

The system performed reliably under normal operating conditions.

---

8. Testing Tools Used

Tool	Purpose
Python	Data testing and preprocessing
Pandas	Dataset validation
SQL	Database testing
Jupyter Notebook	Analytical testing
Power BI	Dashboard testing

---

9. Testing Outcome

The testing process confirmed that:

- The dataset was processed correctly
- SQL queries generated accurate outputs
- Dashboard visualizations matched analytical results
- System modules integrated successfully
- The application performed efficiently

---

10. Conclusion

The Delhi Weather & AQI Analysis System was successfully tested using multiple testing techniques including unit testing, integration testing, database testing, and visualization testing. The testing process ensured the accuracy, reliability, and performance of the system. All modules functioned correctly, and the generated dashboard provided meaningful and accurate environmental insights based on the AQI and weather dataset.

Chapter 7: Results & Discussion

1. Introduction

The Results and Discussion section presents the analytical findings obtained from the Delhi Weather & AQI dataset using Python, SQL, and Power BI visualization techniques. The analysis focused on understanding pollution levels, weather patterns, pollutant concentration, and environmental trends in Delhi during 2025.

Different statistical and visualization methods were used to interpret the dataset and identify meaningful environmental insights.

---

2. Overall AQI Analysis

Average AQI Level

The dashboard analysis showed an average AQI value of approximately **287.36**, indicating that Delhi experienced poor to very poor air quality conditions during most periods of observation.

Discussion

- AQI values above 200 are considered harmful for human health.
- The high AQI reflects severe environmental pollution in urban areas.
- Increased traffic density, industrial emissions, and construction activities are major contributing factors.

---

3. Unhealthy AQI Percentage

The analysis revealed that nearly **77.16%** of AQI observations were classified as unhealthy.

Discussion

- A large portion of the population is exposed to polluted air.
- Prolonged exposure can cause respiratory and cardiovascular diseases.
- The results indicate persistent environmental degradation rather than occasional pollution spikes.

---

4. Pollutant Concentration Analysis

PM2.5 Analysis

The average PM2.5 concentration was approximately **89.54**, which exceeds recommended safety limits.

Discussion

PM2.5 particles are extremely harmful because they can penetrate deep into the lungs and bloodstream. Major sources include:

- Vehicle emissions

- Industrial smoke
- Biomass burning
- Road dust

PM10 Analysis

The dashboard showed an average PM10 level of approximately **300.46**.

Discussion

The extremely high PM10 values indicate:

- Heavy dust pollution
- Construction-related emissions
- Urban particulate matter accumulation

PM10 contributes significantly to poor air quality in Delhi.

---

5. Location-wise AQI Analysis

The analysis identified several pollution hotspots such as:

- Anand Vihar
- Okhla Phase III
- IGI Airport
- Dwarka

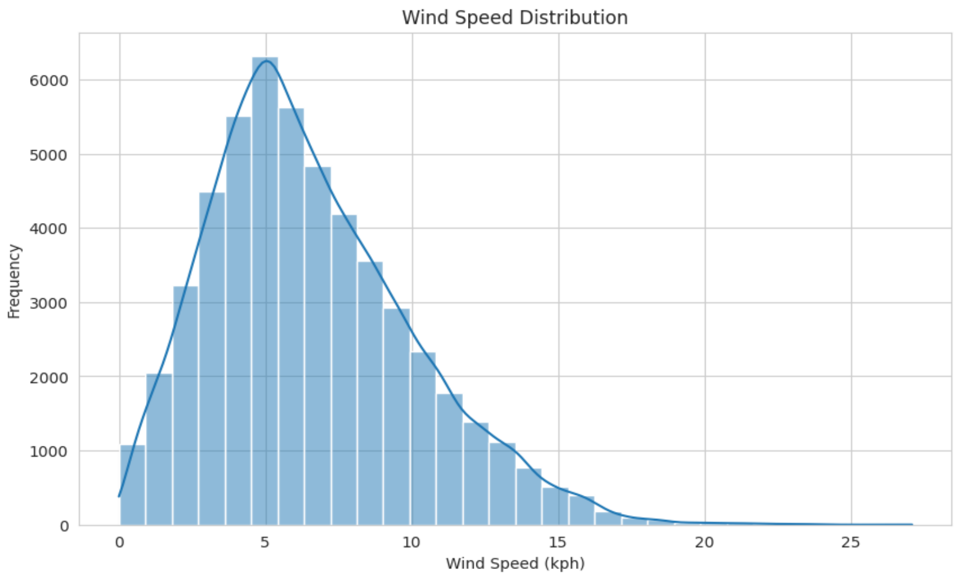
These locations recorded very high AQI and CO concentrations.

Discussion

- Industrial and traffic-dense areas showed higher pollution levels.
- Airport and commercial zones contributed significantly to carbon emissions.
- Anand Vihar emerged as one of the most polluted areas due to heavy transportation activities.

---

5. Wind Speed and AQI Relationship



The dashboard categorized wind conditions into:

- Low Wind
- Medium Wind
- High Wind

Interestingly, higher AQI values were observed during high wind conditions.

Discussion

This observation may occur due to:

- Dust storms
- Regional transport of pollutants
- Seasonal atmospheric disturbances

Although wind usually disperses pollutants, strong winds can also carry suspended dust particles into urban areas.

---

7. Time-Based AQI Trend Analysis

The AQI trend graph indicated that pollution levels increased during evening hours, especially around 6 PM.

Discussion

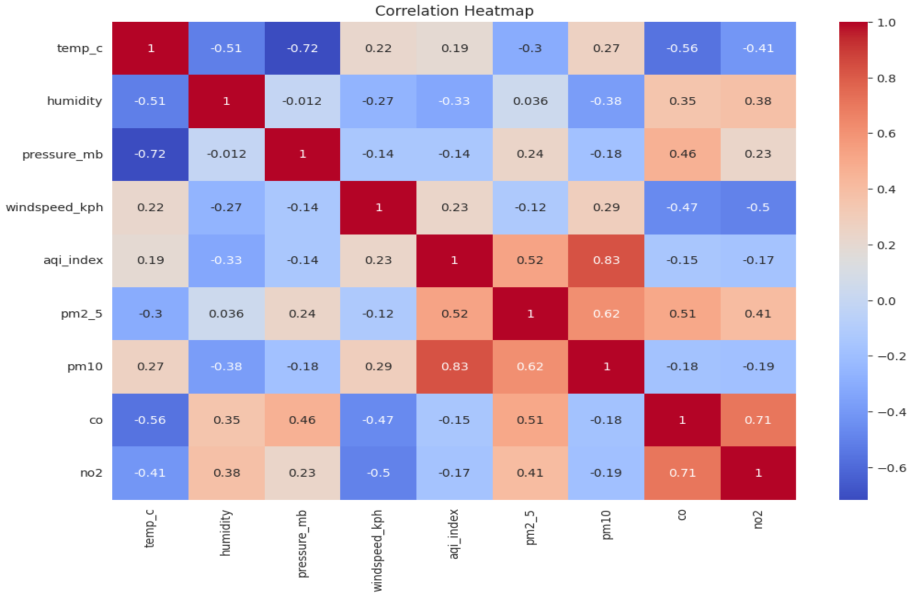
The rise in AQI during evening periods may be caused by:

- Peak traffic congestion
- Industrial operations
- Reduced atmospheric dispersion after sunset

This trend highlights the effect of human activities on urban air quality.

---

8. Correlation Analysis



The correlation heatmap showed strong positive relationships between:

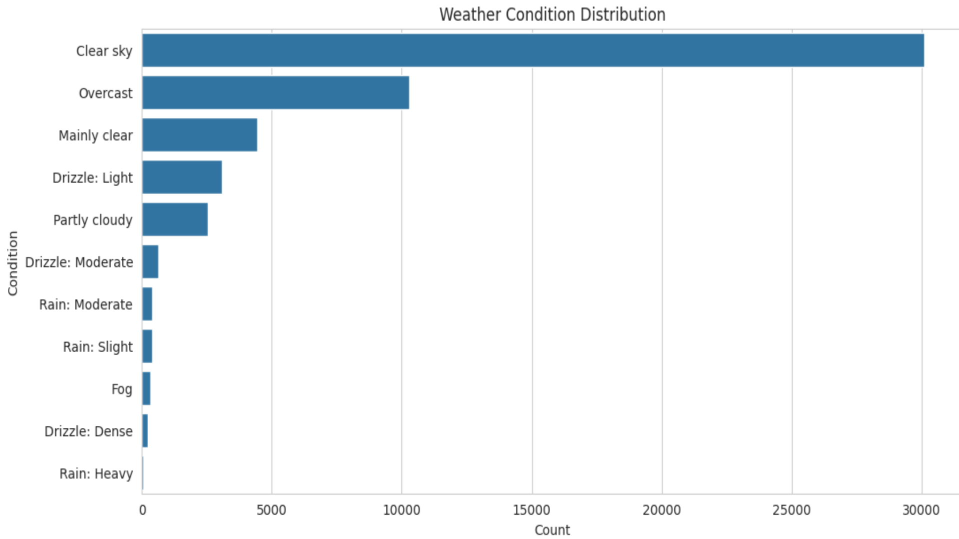
- AQI and PM2.5
- AQI and PM10
- AQI and CO

Discussion

The results confirm that particulate matter and gaseous pollutants directly influence AQI levels. PM2.5 and PM10 were found to be major indicators of poor air quality.

---

10. Weather Condition Analysis

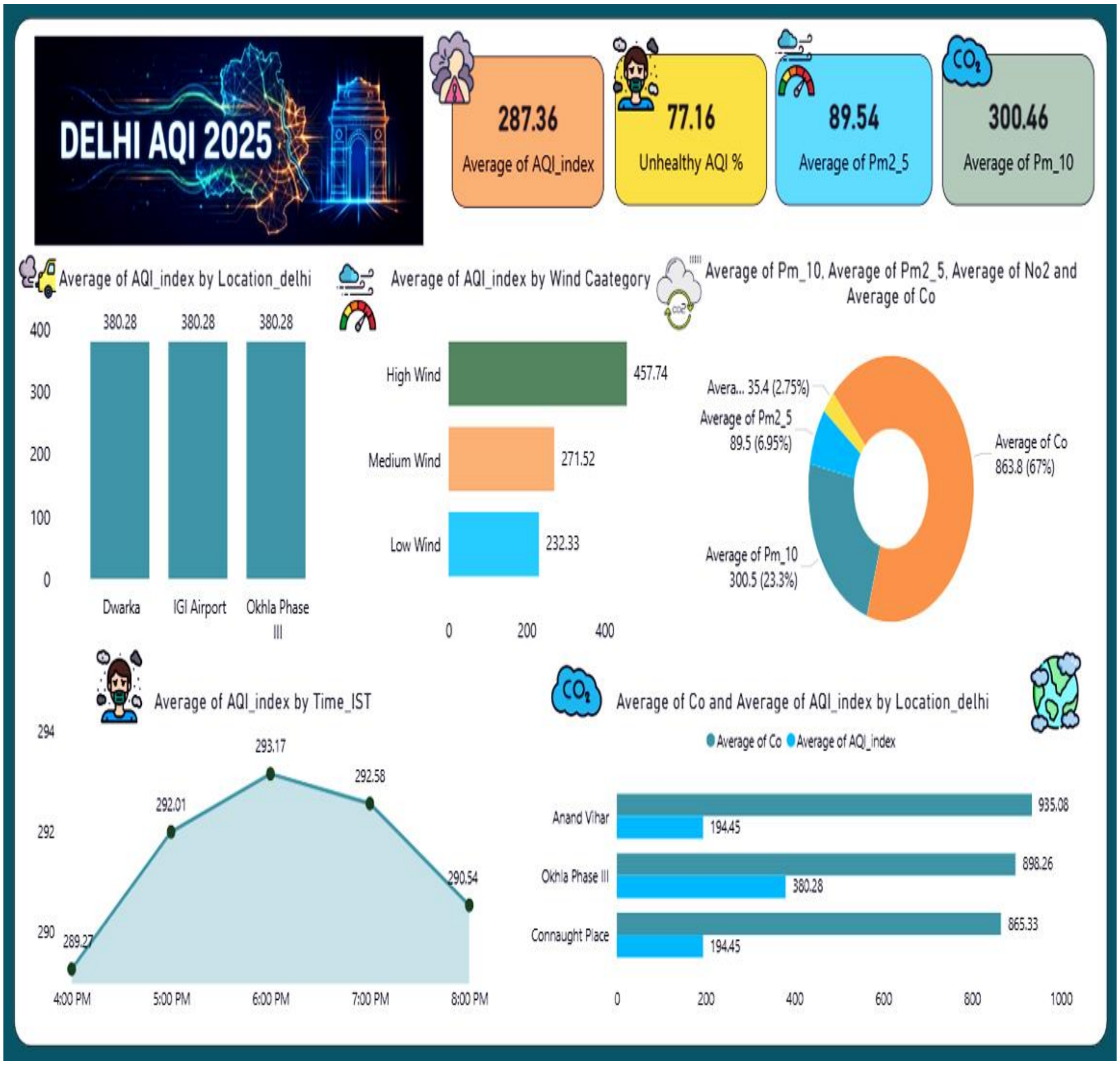


The visualization of weather conditions revealed seasonal variation in pollution levels.

Discussion

- Winter seasons experienced the worst AQI conditions due to fog and low air circulation.
- Monsoon seasons showed improved air quality because rainfall reduced airborne pollutants.
- Temperature and humidity influenced pollutant concentration patterns.

Dashboard Performance and Visualization Results

1. 

The Power BI dashboard successfully displayed:

- KPI metrics
- Pollution trends
- Location-based comparisons
- Pollutant contribution charts
- Time-series AQI analysis

Discussion

The dashboard provided an interactive and user-friendly way to analyze complex environmental datasets. Visualization techniques improved understanding of pollution patterns and environmental risks.

---

11. Overall Findings

The study revealed that:

- Delhi experiences critically high pollution levels.
- PM2.5 and PM10 are the primary contributors to AQI.
- Traffic and industrial activities strongly affect air quality.
- Seasonal weather changes influence pollutant concentration.
- Environmental monitoring systems are essential for pollution management.

---

12. Conclusion

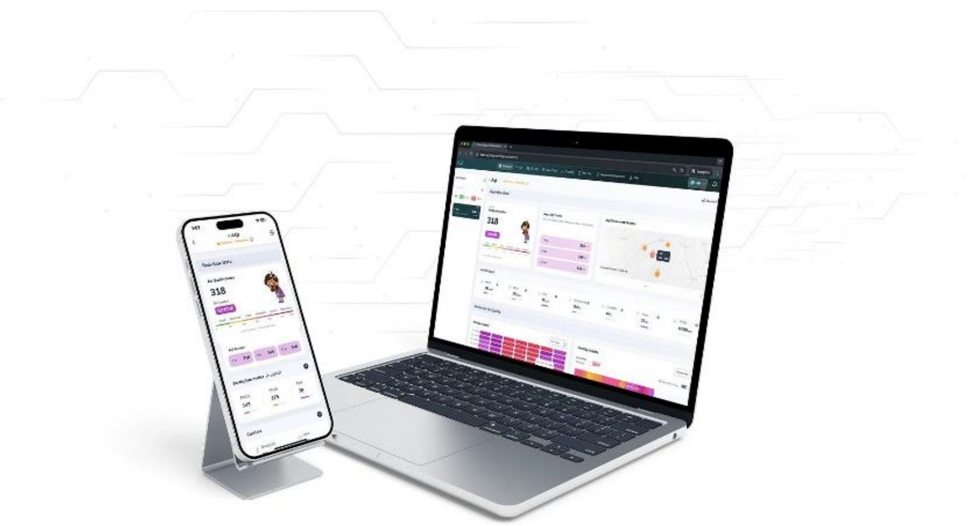
The results obtained from the Delhi Weather & AQI Analysis System demonstrate the effectiveness of data analytics, SQL querying, and dashboard visualization in environmental monitoring. The project successfully identified pollution hotspots, pollutant trends, and weather-related AQI variations in Delhi.

The findings emphasize the urgent need for:

- Pollution control strategies
- Improved environmental policies
- Real-time AQI monitoring systems
- Sustainable urban planning

The project also demonstrates the practical application of Python, SQL, and Power BI in environmental data analysis and visualization.

Chapter 8 : Conclusion & Future Scope



The Delhi Weather & AQI Analysis System was successfully developed to analyze and visualize environmental and air pollution data using Python, SQL, and Power BI technologies. The project effectively processed the Delhi AQI and weather dataset, performed data cleaning and analysis, and generated meaningful visual insights regarding pollution trends and atmospheric conditions.

The study revealed that Delhi experiences consistently high pollution levels, with AQI values frequently falling under poor and unhealthy categories. Pollutants such as PM2.5 and PM10 were identified as the major contributors to deteriorating air quality. The analysis also showed that weather conditions such as wind speed, humidity, and seasonal changes significantly affect pollution concentration levels.

Using SQL queries and interactive dashboard visualizations, the system successfully identified:

- Pollution hotspots
- Monthly and hourly AQI trends
- Pollutant contribution patterns
- Weather and AQI relationships
- Environmental risk indicators

The Power BI dashboard provided an interactive and user-friendly interface for monitoring and interpreting environmental data. The project demonstrates the practical application of:

- Data preprocessing
- SQL analytics
- Data visualization
- Dashboard development
- Environmental data analysis

Overall, the system provides an effective framework for AQI monitoring and environmental analytics, helping users better understand pollution patterns and supporting awareness regarding urban environmental conditions.

Future Scope

The project can be further enhanced with advanced technologies and real-time monitoring capabilities. Several future improvements can make the system more intelligent, scalable, and useful for environmental management.

1. Real-Time Data Integration

The current system uses a static CSV dataset. In the future, real-time AQI and weather APIs can be integrated to provide live environmental monitoring and instant updates.

Possible APIs

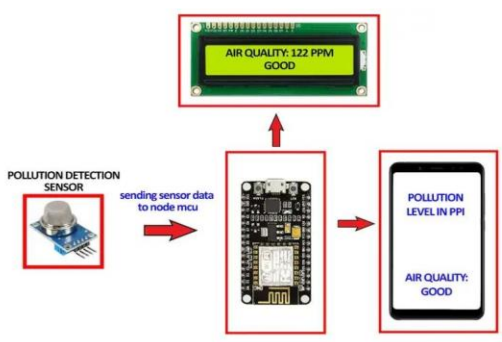
- OpenWeather API
- AQICN API
- Government pollution monitoring APIs

---

2. Machine Learning Prediction Models

Machine learning algorithms can be added to:

- Predict future AQI levels
- Forecast pollution trends
- Detect abnormal pollution spikes



Algorithms

- Linear Regression
- Random Forest
- ARIMA Time Series
- LSTM Neural Networks

---

3. IoT Sensor Integration

IoT-based air quality sensors can be connected to the system for live environmental data collection from multiple locations across Delhi.

Benefits

- Real-time pollution tracking
- Smart city monitoring
- Faster environmental alerts

---

4. Geographic Heatmap Visualization

GIS and map-based visualization can be added to display pollution intensity across Delhi regions using:

- Heatmaps
- Interactive maps
- Geospatial analytics

---

5. Mobile Application Development

A mobile application can be developed to:

- Display live AQI
- Send pollution alerts
- Provide health recommendations
- Track nearby pollution levels

---

6. Automated Alert System

The system can automatically notify users when pollution levels exceed dangerous thresholds through:

- SMS alerts
- Email notifications
- Mobile push notifications

---

7. Advanced Dashboard Features

Future dashboards can include:

- Drill-down analytics
- Dynamic filtering
- Real-time streaming dashboards
- AI-generated environmental insights

---

8. Government and Public Health Applications

The system can support:

- Environmental policy planning
- Urban pollution management
- Public health monitoring
- Smart city initiatives

Final Remark

The Delhi Weather & AQI Analysis System demonstrates how modern data analytics and visualization technologies can be used to study environmental pollution effectively. With future integration of AI, IoT, and real-time monitoring systems, the project can evolve into a smart environmental intelligence platform for sustainable urban development and pollution control.

Chapter 9: References

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Chapter 10: Appendices

Dataset Loading

```
import pandas as pd

df = pd.read_csv("delhi-weather-aqi-2025.csv")

print(df.head())
```

Data Preprocessing

```
df['date_ist'] = pd.to_datetime(df['date_ist'])

df.dropna(inplace=True)

df['Month'] = df['date_ist'].dt.month
df['Year'] = df['date_ist'].dt.year
```

AQI Trend Visualization

```
import matplotlib.pyplot as plt

monthly_aqi = df.groupby('Month')['aqi_index'].mean()

plt.plot(monthly_aqi.index, monthly_aqi.values)
plt.title("Monthly AQI Trend")
plt.xlabel("Month")
plt.ylabel("Average AQI")
plt.show()
```

Correlation Heatmap

```
import seaborn as sns

sns.heatmap(df.corr(), annot=True)
plt.show()
```

Appendix B – Sample SQL Queries

Create AQI Table

```
CREATE TABLE aqi (
date_ist DATE,
location_delhi VARCHAR(100),
aqi_index FLOAT,
pm2_5 FLOAT,
pm10 FLOAT,
co FLOAT,
no2 FLOAT,
temp_c FLOAT,
humidity FLOAT,
windspeed_kph FLOAT
);
```

Average AQI Query

```
SELECT AVG(aqi_index) AS average_aqi
FROM aqi;
```

Top Polluted Locations

```
SELECT location_delhi,
AVG(aqi_index) AS avg_aqi
FROM aqi
GROUP BY location_delhi
ORDER BY avg_aqi DESC
LIMIT 5;
```

Wind Category Analysis

```
SELECT
CASE
WHEN windspeed_kph < 5 THEN 'Low Wind'
WHEN windspeed_kph BETWEEN 5 AND 10 THEN 'Medium Wind'
ELSE 'High Wind'
END AS wind_category,
AVG(aqi_index) AS avg_aqi
FROM aqi
GROUP BY wind_category;
```

Appendix C – Dashboard Screenshots

Dashboard Components

The Power BI dashboard contains:

- KPI Cards
- AQI Trend Graphs
- Pollutant Contribution Charts
- Wind Speed Analysis
- Location-wise AQI Comparison
- Correlation Analysis

Dashboard Insights

- Average AQI: 287.36
- Unhealthy AQI Percentage: 77.16%
- Average PM2.5: 89.54
- Average PM10: 300.46

Appendix D – Dataset Description

Column Name	Description
date_ist	Date of observation
location_delhi	Monitoring location
aqi_index	Air Quality Index value
pm2_5	Fine particulate matter
pm10	Coarse particulate matter
co	Carbon monoxide level

Column Name	Description
no2	Nitrogen dioxide level
temp_c	Temperature in Celsius
humidity	Relative humidity
windspeed_kph	Wind speed in kilometers/hour

Appendix E – Software and Tools Used

Software/Tool	Purpose
Python	Data analysis
Pandas	Data preprocessing
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical plotting
SQL	Database querying
PostgreSQL/MySQL	Database management
Power BI	Dashboard development
Jupyter Notebook	Development environment

Appendix F – Hardware and Software Requirements

Hardware Requirements

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 10 GB free space

Software Requirements

- Python 3.x
- PostgreSQL/MySQL
- Jupyter Notebook
- Power BI Desktop
- VS Code / PyCharm

Appendix G – Output Screens

Generated Outputs

- AQI Dashboard
- Monthly AQI Trend Graph
- Temperature Distribution Graph
- Correlation Heatmap
- Pollution Hotspot Analysis
- SQL Query Outputs

Appendix H – Project Workflow

```
Dataset Collection
↓
Data Cleaning
↓
Feature Engineering
↓
SQL Database Storage
↓
SQL Query Analysis
↓
Data Visualization
↓
Power BI Dashboard
↓
Result Interpretation
```